

Cloud Networking Fundamentals: Complete Documentation

VNet Peering & Multi-Tier Application Communication

Table of Contents

- 1. [OSI Model Fundamentals](#)
- 2. [Layer 3 and Layer 4 Communication](#)
- 3. [VLAN Concepts](#)
- 4. [Routing Fundamentals](#)
- 5. [Network Security Groups \(NSGs\)](#)
- 6. [Virtual Networks \(VNETs\)](#)
- 7. [VNet Peering](#)
- 8. [Transitive Peering](#)
- 9. [2-Tier Application Architecture](#)
- 10. [3-Tier Application Architecture](#)
- 11. [Implementation Process](#)

Section 1: OSI Model Fundamentals

1.1 What is the OSI Model?

The **Open Systems Interconnection (OSI) model** is a conceptual framework that standardizes how different network systems communicate with each other. It divides network communication into **seven distinct layers**, each with specific responsibilities.

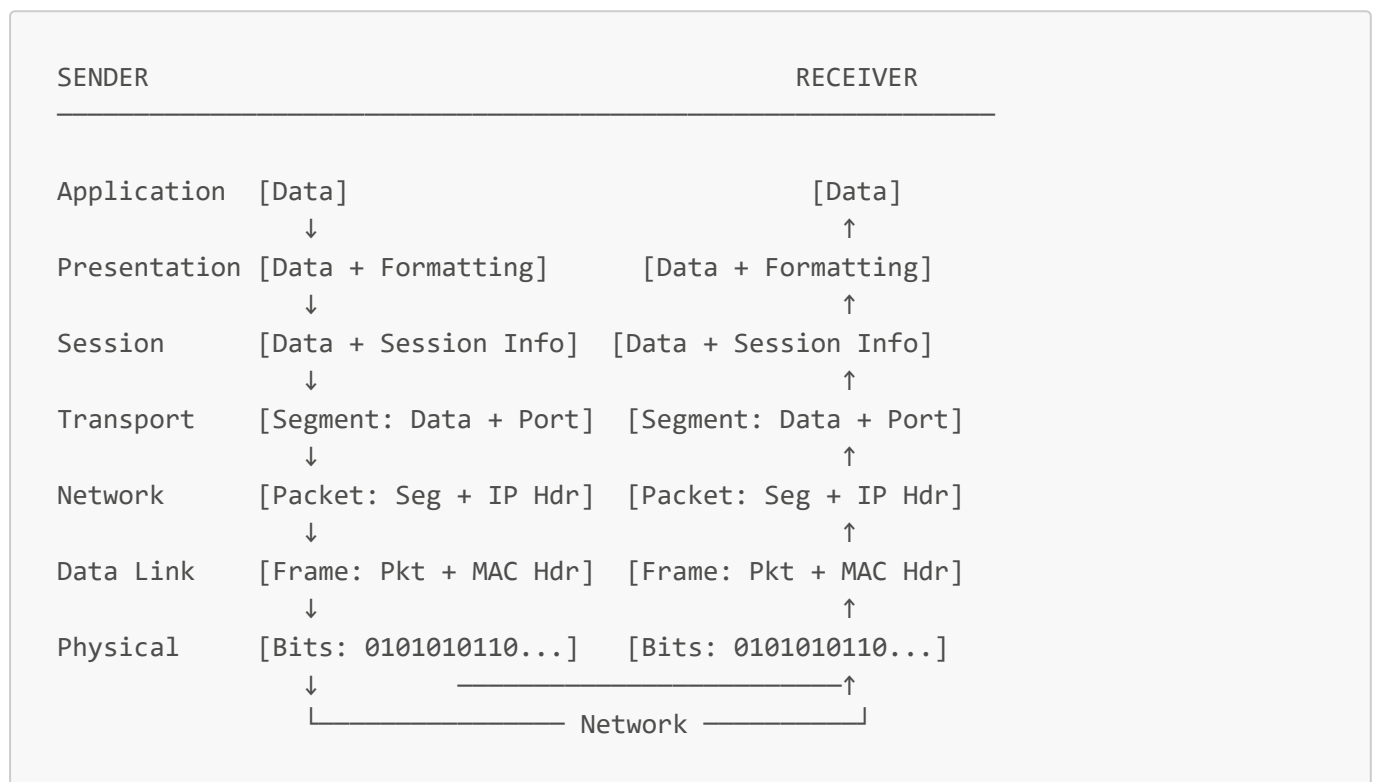
Think of the OSI model as a postal system — each layer handles a specific part of getting your message from sender to receiver, and each layer only communicates with the layers directly above and below it.

OSI MODEL LAYERS		
Layer	Name	Responsibility
7	Application	End-user interface, HTTP, FTP, DNS
6	Presentation	Data formatting, encryption, SSL/TLS
5	Session	Session management, authentication
4	Transport	End-to-end delivery, TCP/UDP, ports

3	Network	Logical addressing, routing, IP
2	Data Link	MAC addressing, frames, switches
1	Physical	Cables, signals, bits, hardware

1.2 How Data Flows Through OSI Layers

When you send data, it travels **down** through the layers on the sender's side (encapsulation) and **up** through the layers on the receiver's side (decapsulation).



1.3 Data Units at Each Layer

Layer 7 - Application	Data / Message
Layer 6 - Presentation	Data
Layer 5 - Session	Data
Layer 4 - Transport	Segment
Layer 3 - Network	Packet
Layer 2 - Data Link	Frame
Layer 1 - Physical	Bits

1.4 Why OSI Model Matters in Cloud Networking

Cloud Concept	OSI Layer	Relevance
Virtual Networks (VNet)	Layer 3	IP-based logical segmentation
NSG Rules	Layer 3 & 4	IP addresses and port filtering
VNet Peering	Layer 3	Routing between networks
Load Balancers	Layer 4 & 7	Port and application routing
VLANs	Layer 2	Network segmentation
SSL/TLS Certificates	Layer 6	Encryption

Section 2: Layer 3 and Layer 4 Communication

2.1 Layer 3 — The Network Layer

What is Layer 3?

Layer 3 is responsible for **logical addressing** and **routing** — it determines how packets move from one network to another across multiple hops.

Key Components of Layer 3

IP Addressing

Every device on a network receives a unique **IP address** that identifies it logically.

IPv4 Address Structure:

192.168.10.50 / 24

Prefix Length (Subnet Mask)

Host Portion (identifies device)

Network Portion (identifies network)

Binary Representation:

11000000.10101000.00001010.00110010

— Network (24 bits) —

— Host (8 bits) —

Subnet Masks

Common Subnet Masks

CIDR	Subnet Mask	Hosts	Use Case
/8	255.0.0.0	16M+	Large Corp
/16	255.255.0.0	65,534	Large VNet
/24	255.255.255.0	254	Subnet
/26	255.255.255.192	62	Small Subnet
/28	255.255.255.240	14	Tiny Subnet
/32	255.255.255.255	1	Single Host

How Layer 3 Routing Works

Source: 10.1.0.5 → Destination: 10.2.0.10

Step 1: Source checks if destination is on same network
10.1.0.5/24 → Network: 10.1.0.0
10.2.0.10/24 → Network: 10.2.0.0
Different networks! → Send to Default Gateway

Step 2: Default Gateway (Router) checks routing table

ROUTING TABLE	
Destination	Next Hop
10.1.0.0/24	Directly Connected
10.2.0.0/24	10.0.0.1 (Next Router)
0.0.0.0/0	Internet Gateway

Step 3: Packet forwarded to next hop until destination is reached

2.2 Layer 4 — The Transport Layer

What is Layer 4?

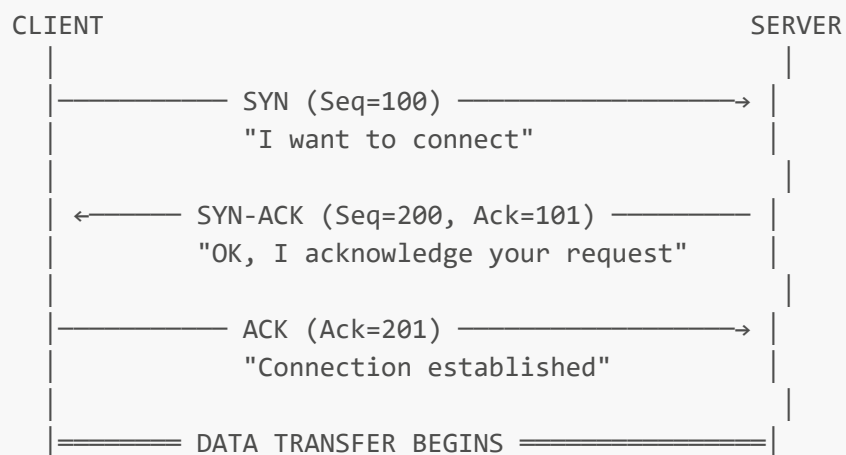
Layer 4 is responsible for **end-to-end communication** between applications. It uses **port numbers** to identify specific applications and services on a device.

TCP vs UDP

TCP vs UDP Comparison	
TCP	UDP
Connection-oriented	Connectionless

Reliable delivery	No guaranteed delivery
Ordered packets	No ordering
Error checking	Minimal error checking
Slower (overhead)	Faster (low overhead)
HTTP, HTTPS, SSH, FTP	DNS, VoIP, Video Streaming

TCP Three-Way Handshake



Port Numbers

Well-Known Port Numbers		
Port	Protocol	Service
22	TCP	SSH (Secure Shell)
80	TCP	HTTP (Web Traffic)
443	TCP	HTTPS (Secure Web)
3389	TCP	RDP (Remote Desktop)
1433	TCP	SQL Server
3306	TCP	MySQL
5432	TCP	PostgreSQL
6379	TCP	Redis
53	UDP	DNS
8080	TCP	HTTP Alternate

Layer 4 in Cloud Context (NSG Rules)

NSG Rule Example:

Rule: Allow HTTPS Traffic

Source IP:	0.0.0.0/0	(Any)
Source Port:	*	(Any)
Destination IP:	10.1.0.4	(Web Server)
Destination Port:	443	(HTTPS - Layer 4)
Protocol:	TCP	(Layer 4)
Action:	Allow	

This rule operates at Layer 3 (IP) and Layer 4 (Port)

Section 3: VLAN Concepts

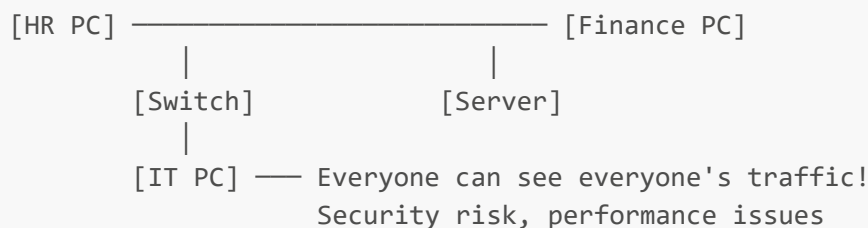
3.1 What is a VLAN?

A **Virtual Local Area Network (VLAN)** is a logical grouping of network devices that operates as if they are on the same physical network, even if they are physically spread across different locations.

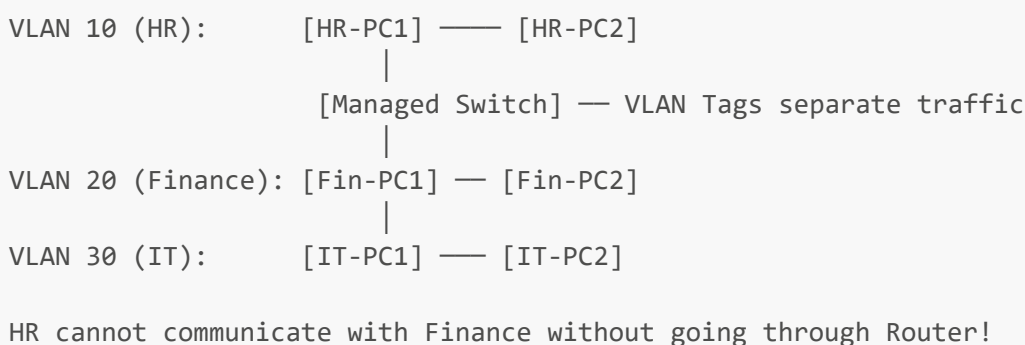
VLANs operate at **Layer 2 (Data Link Layer)** and use **VLAN tags (802.1Q)** to identify which VLAN a frame belongs to.

3.2 Why VLANs are Used

WITHOUT VLANs (Flat Network):



WITH VLANs (Segmented Network):



3.3 How VLAN Tagging Works (802.1Q)

Standard Ethernet Frame:

Dest MAC	Src MAC	EtherType	Data
(6 bytes)	(6 bytes)	(2 bytes)	

802.1Q Tagged Frame:

Dest MAC	Src MAC	802.1Q Tag	EtherType	Data
(6 bytes)	(6 bytes)	(4 bytes)	(2 bytes)	



VLAN Tag	
TPID: 0x8100	← Identifies as 802.1Q
PCP: 3 bits	← Priority
DEI: 1 bit	← Drop Eligible
VID: 12 bits	← VLAN ID (1-4094)

3.4 VLANs in Cloud Computing

In cloud environments like **Azure**, the concept of VLANs translates into **Subnets within Virtual Networks (VNETs)**:

Traditional Network		Cloud Equivalent
Physical Network	→	Virtual Network (VNet)
VLAN	→	Subnet
VLAN ID	→	Subnet CIDR Range
Inter-VLAN Routing	→	VNet Routing / Peering
VLAN ACLs	→	Network Security Groups (NSGs)

Azure VNet with Subnets (Equivalent to VLANs)

```

Azure Virtual Network: 10.0.0.0/16
├── Subnet: Web-Tier    (10.0.1.0/24) → Like VLAN 10
│   └── Web Servers (10.0.1.4, 10.0.1.5)
└── Subnet: App-Tier   (10.0.2.0/24) → Like VLAN 20
  
```

```
├── App Servers (10.0.2.4, 10.0.2.5)
└── Subnet: DB-Tier (10.0.3.0/24) → Like VLAN 30
    ├── DB Servers (10.0.3.4, 10.0.3.5)
```

Section 4: Routing Fundamentals

4.1 What is Routing?

Routing is the process of selecting paths in a network along which to send network traffic. A **router** examines each incoming packet's destination IP address and forwards it to the appropriate next-hop location.

4.2 Types of Routing

Types of Routing
STATIC ROUTING • Manually configured routes • No automatic updates • Simple, predictable • Good for small/stable networks DYNAMIC ROUTING • Routes learned automatically via protocols • Protocols: BGP, OSPF, RIP • Adapts to network changes • Used in large, complex networks SYSTEM ROUTES (Azure) • Automatically created by Azure • Handles VNet, Subnet, VNet Peering traffic • Cannot be deleted but can be overridden

4.3 Routing Table

A routing table is a database of routes that tells the router where to send packets.

ROUTING TABLE EXAMPLE:

Destination	Subnet Mask	Next Hop	Metric
10.1.0.0	255.255.255.0	On-Link	1
10.2.0.0	255.255.255.0	10.1.0.1	10
10.3.0.0	255.255.255.0	10.1.0.1	10
0.0.0.0	0.0.0.0	10.1.0.1	1

Reading the table:

- 10.1.0.0/24 → Directly connected (On-Link)
- 10.2.0.0/24 → Send to 10.1.0.1 (next hop router)
- 10.3.0.0/24 → Send to 10.1.0.1 (next hop router)
- 0.0.0.0/0 → Default route (everything else)

4.4 Azure System Routes

Azure automatically creates system routes for each subnet:

Azure Auto-Created Routes:

VNet: 10.0.0.0/16

Subnet: 10.0.1.0/24

Address Prefix	Next Hop Type
10.0.0.0/16	Virtual Network (VNet local)
0.0.0.0/0	Internet
10.0.0.0/8	None (RFC 1918 - blocked)
172.16.0.0/12	None (RFC 1918 - blocked)
192.168.0.0/16	None (RFC 1918 - blocked)

When VNet Peering is added:

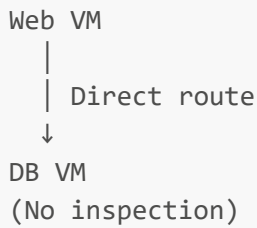
10.2.0.0/16	VNet Peering (peered network)
-------------	-------------------------------

4.5 User-Defined Routes (UDR)

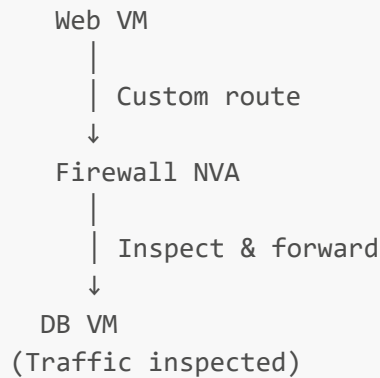
UDRs allow you to override Azure's default system routes:

Use Case: Force traffic through a Network Virtual Appliance (NVA)

Without UDR:



With UDR:



UDR Configuration:

Address Prefix: 10.0.3.0/24 (DB Subnet)
Next Hop Type: Virtual Appliance
Next Hop IP: 10.0.4.4 (Firewall IP)

Section 5: Network Security Groups (NSGs)

5.1 What is an NSG?

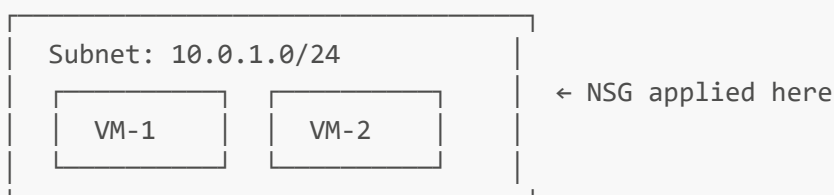
A **Network Security Group (NSG)** is a cloud firewall that contains a list of security rules that **allow or deny** inbound and outbound network traffic based on:

- Source IP address / CIDR range
- Destination IP address / CIDR range
- Protocol (TCP, UDP, Any)
- Port numbers (source and destination)
- Direction (Inbound or Outbound)

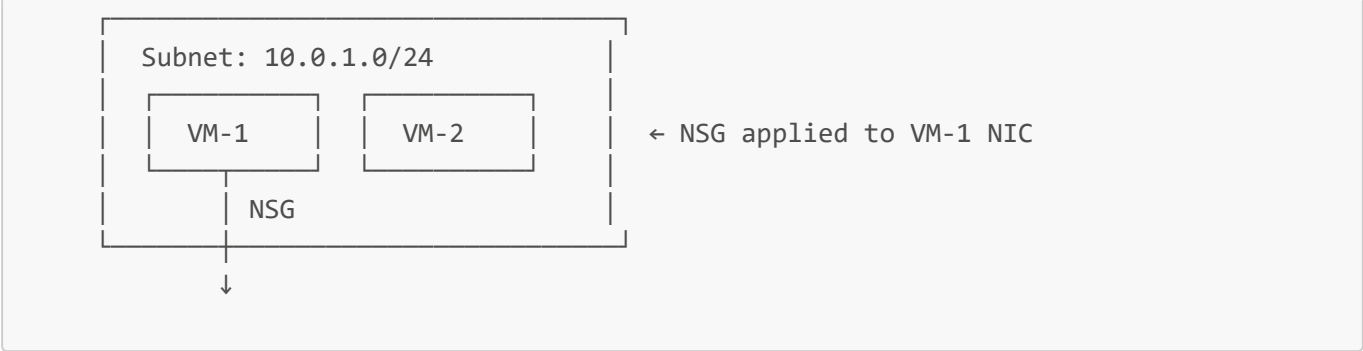
5.2 NSG Architecture

NSG can be associated with:

1. SUBNET LEVEL (Applies to all resources in subnet)



2. NIC (NETWORK INTERFACE CARD) LEVEL (Applies to single VM)



5.3 NSG Rule Structure

NSG RULE COMPONENTS:

NSG Rule	
Priority	100 - 4096 (lower = higher priority, processed first)
Name	Descriptive name
Direction	Inbound or Outbound
Source	IP, CIDR, Service Tag, ASG
Source Port Range	*, single port, or range
Destination	IP, CIDR, Service Tag, ASG
Destination Port	*, single port, or range
Protocol	TCP, UDP, ICMP, Any
Action	Allow or Deny

5.4 Default NSG Rules

Azure includes default rules that cannot be deleted:

DEFAULT INBOUND RULES:

Priority	Name	Source	Port	Action
65000	AllowVnetInBound	VirtualNetwork	*	Allow
65001	AllowAzureLBIn	AzureLoadBal	*	Allow
65500	DenyAllInBound	*	*	Deny

DEFAULT OUTBOUND RULES:

Priority	Name	Dest	Port	Action
65000	AllowVnetOutBound	VNet	*	Allow
65001	AllowInternetOut	Inet	*	Allow
65500	DenyAllOutBound	*	*	Deny

5.5 NSG Traffic Flow Processing

INBOUND TRAFFIC PROCESSING:

Internet → Subnet NSG → NIC NSG → Virtual Machine

- Step 1: Packet arrives at Azure border
- Step 2: Subnet-level NSG rules evaluated (lowest priority first)
- Step 3: If allowed, NIC-level NSG rules evaluated
- Step 4: If allowed, packet reaches the VM

OUTBOUND TRAFFIC PROCESSING:

Virtual Machine → NIC NSG → Subnet NSG → Internet

- Step 1: Packet leaves VM
- Step 2: NIC-level NSG rules evaluated
- Step 3: If allowed, Subnet-level NSG rules evaluated
- Step 4: If allowed, packet exits to destination

5.6 NSG Service Tags

Service Tags are predefined identifiers for Azure services, simplifying NSG configuration:

Service Tag	Represents
Internet	Public Internet space
VirtualNetwork	All VNet address spaces
AzureLoadBalancer	Azure Load Balancer IPs
AzureCloud	All Azure datacenter IPs
Storage	Azure Storage service IPs
Sql	Azure SQL service IPs
AppService	Azure App Service IPs

Usage in NSG Rule:

Source: Internet → Destination: 10.0.1.0/24 Port: 443 ALLOW
(Instead of specifying all internet IPs)

5.7 NSG for 3-Tier Application

3-TIER NSG DESIGN:

WEB TIER NSG (Subnet: 10.0.1.0/24):

Priority	Direction	Source	Port	Action
100	Inbound	Internet	80	Allow
110	Inbound	Internet	443	Allow
200	Outbound	10.0.1.0/24	8080	Allow
65500	Inbound	*	*	Deny

APP TIER NSG (Subnet: 10.0.2.0/24):

Priority	Direction	Source	Port	Action
100	Inbound	10.0.1.0/24	8080	Allow
200	Outbound	10.0.2.0/24	1433	Allow
65500	Inbound	*	*	Deny

DB TIER NSG (Subnet: 10.0.3.0/24):

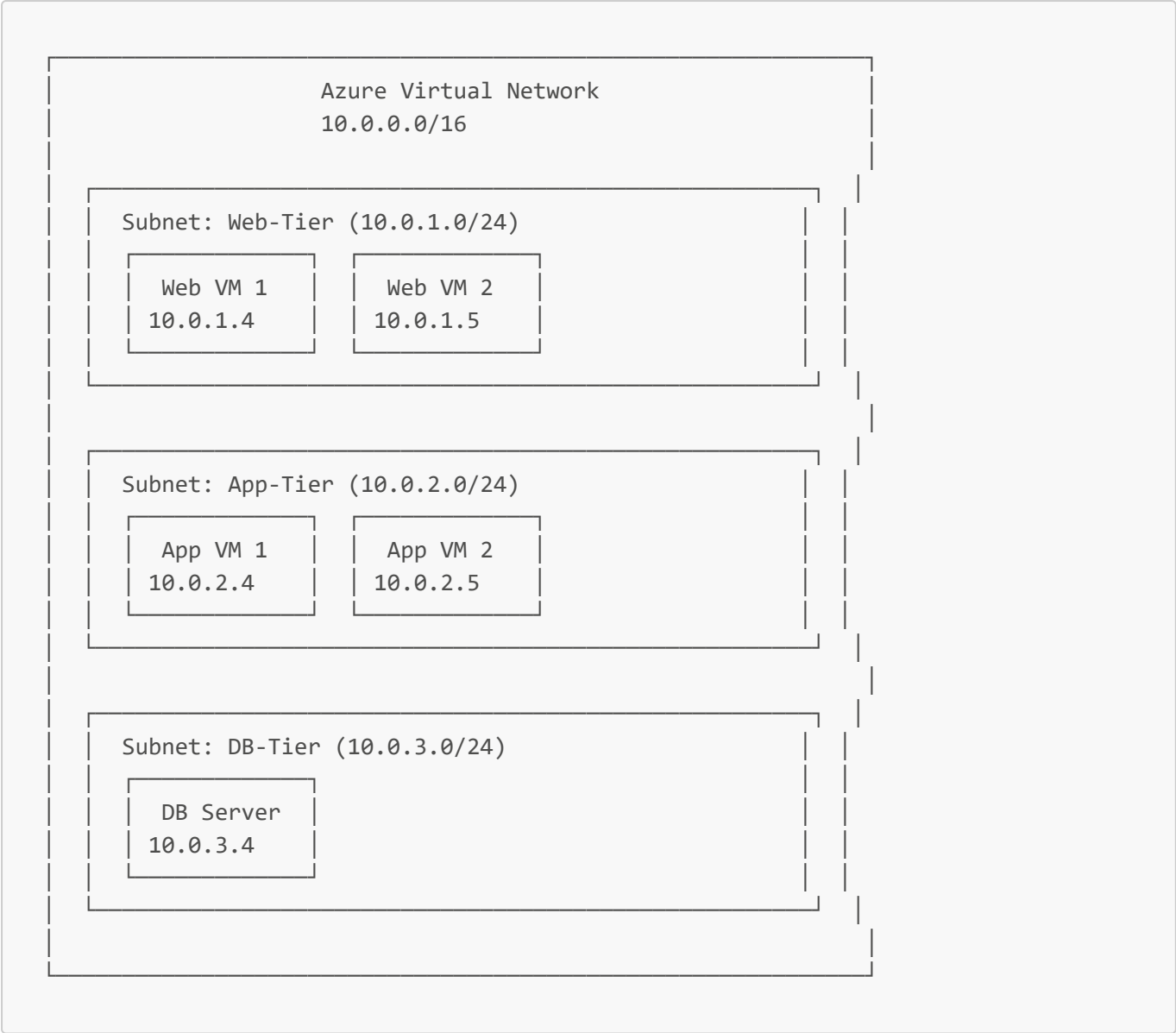
Priority	Direction	Source	Port	Action
100	Inbound	10.0.2.0/24	1433	Allow
65500	Inbound	*	*	Deny

Section 6: Virtual Networks (VNETs)

6.1 What is a Virtual Network?

A **Virtual Network (VNet)** is a logically isolated network in the cloud that closely resembles a traditional network you'd operate in your own data center. It enables Azure resources to securely communicate with each other, the internet, and on-premises networks.

6.2 VNet Components



6.3 VNet Address Space Planning

ADDRESS SPACE PLANNING GUIDELINES:

RFC 1918 Private Address Ranges (Recommended for VNets):

Range	Size
10.0.0.0/8	16,777,216 addresses
172.16.0.0/12	1,048,576 addresses
192.168.0.0/16	65,536 addresses

AZURE RESERVED ADDRESSES IN EACH SUBNET:
Azure reserves 5 IP addresses in every subnet:

Subnet: 10.0.1.0/24

10.0.1.0

→ Network Address

10.0.1.1

→ Default Gateway (Azure)

- 10.0.1.2 → Azure DNS
- 10.0.1.3 → Azure Reserved
- 10.0.1.255 → Broadcast Address

Usable addresses: 10.0.1.4 → 10.0.1.254 = 251 addresses

6.4 VNet Key Properties

VNet Key Properties	
Property	Description
Region-Specific	VNet exists in one Azure region
Subscription-Scoped	Belongs to one subscription
Multiple CIDR Blocks	Can have multiple address spaces
DNS Configuration	Custom or Azure-provided DNS
DDoS Protection	Basic (free) or Standard (paid)
No Cross-Region	Cannot span multiple regions

Section 7: VNet Peering

7.1 What is VNet Peering?

VNet Peering connects two Azure virtual networks, enabling resources in either virtual network to communicate with each other. The traffic between peered VNets travels through Microsoft's private backbone network — **not** through the public internet.

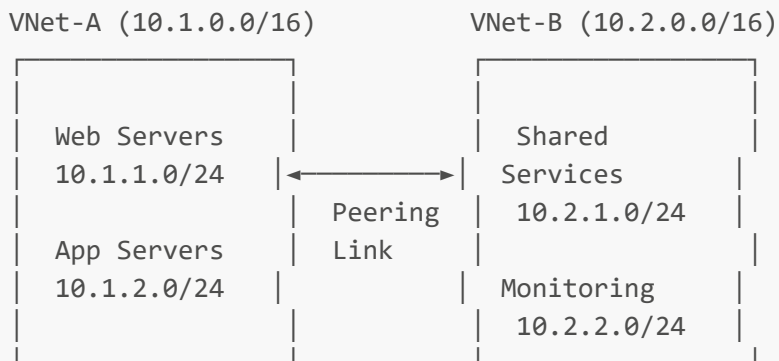
7.2 Types of VNet Peering

Types of VNet Peering
1. VIRTUAL NETWORK PEERING • Connects VNets within the SAME region • Ultra-low latency • High bandwidth • Most common type 2. GLOBAL VIRTUAL NETWORK PEERING • Connects VNets in DIFFERENT regions • Slightly higher latency (cross-region)

- Traffic stays on Microsoft backbone
- Useful for DR and multi-region architectures

7.3 VNet Peering Architecture

VNet Peering Connection:



- Both directions must be configured (bi-directional)
- Non-overlapping address spaces required
- No gateway required for peering
- Traffic stays on Azure backbone

7.4 VNet Peering Requirements

REQUIREMENTS FOR VNET PEERING:

✓ REQUIREMENTS:

1. Non-overlapping address spaces
 VNet-A: 10.1.0.0/16 ✓
 VNet-B: 10.2.0.0/16 ✓
 (Cannot use 10.0.0.0/16 for both)
2. Peering is not transitive
 A peers B, B peers C ≠ A can reach C
3. Both peerings must be created
 A→B peering AND B→A peering required
4. Sufficient permissions
 Contributor role on both VNets

✗ LIMITATIONS:

- Overlapping CIDR blocks not allowed
- Cannot resize address space after peering
- Maximum 500 peerings per VNet (default)
- Peering is NOT transitive (requires hub-spoke)

7.5 VNet Peering Configuration Options

PEERING CONFIGURATION SETTINGS:

When creating a peering link, configure:

SETTING	OPTIONS	DESCRIPTION
Traffic to remote VNet	Allow/Block	Forward traffic
Traffic from remote VNet	Allow/Block	Receive traffic
Gateway transit	On/Off	Use peer gateway
Remote gateways	On/Off	Use remote gw

Gateway Transit Use Case:

Hub VNet has VPN/ExpressRoute gateway

Spoke VNets can use Hub's gateway via "Use Remote Gateway"

This avoids creating gateway in each spoke

7.6 How Peering Updates Routing Tables

When peering is established, Azure automatically updates routing tables:

BEFORE PEERING:

VNet-A Route Table:

Destination	Next Hop
10.1.0.0/16	Virtual Network
0.0.0.0/0	Internet

AFTER PEERING VNet-A ↔ VNet-B:

VNet-A Route Table (Updated Automatically):

Destination	Next Hop
10.1.0.0/16	Virtual Network
10.2.0.0/16	VNet Peering ← NEW
0.0.0.0/0	Internet

Section 8: Transitive Peering

8.1 What is Transitive Peering?

Transitive peering refers to the concept of routing traffic through an intermediate VNet to reach another VNet that is not directly peered.

Critical Concept: Azure VNet Peering is **NOT transitive by default**.

8.2 The Non-Transitive Problem

THE PROBLEM – Non-Transitive Peering:

VNet-A ← Peered → VNet-B ← Peered → VNet-C

Question: Can VNet-A communicate with VNet-C?

ANSWER: ❌ NO – Peering is NOT transitive!

Even though:

A peers with B ✅

B peers with C ✅

A CANNOT reach C directly ❌

VNet-A (10.1.0.0/16)

|
| Peering

VNet-B (10.2.0.0/16) ← Hub

|
| Peering

VNet-C (10.3.0.0/16)

VNet-A → VNet-B = ✅ (Allowed)

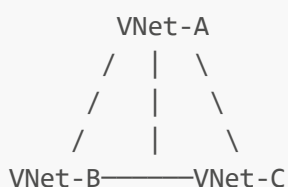
VNet-B → VNet-C = ✅ (Allowed)

VNet-A → VNet-C = ❌ (NOT Allowed – must add direct peering)

8.3 Solutions for Transitive Connectivity

Solution 1: Full Mesh Peering (Small Scale)

FULL MESH – Direct peering between all VNets:



Peerings:

- A ↔ B (2 peering links)
- B ↔ C (2 peering links)
- A ↔ C (2 peering links)

Total: 6 peering links for 3 VNets

Formula: $n*(n-1)$ peering links for n VNets

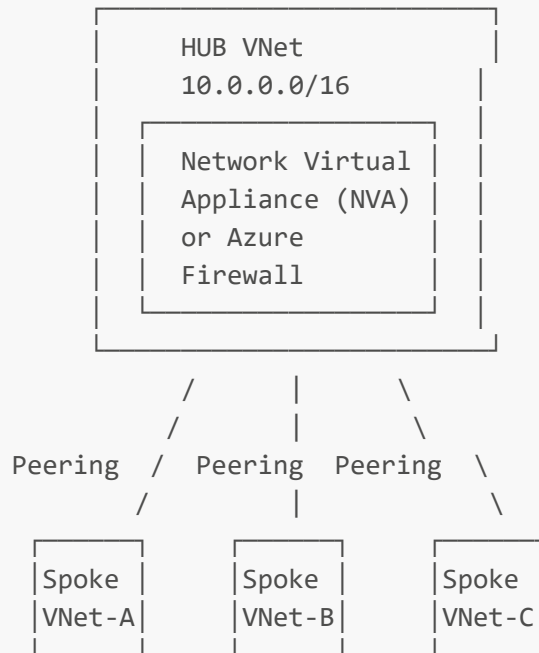
- 4 VNets = 12 links

- 10 VNets = 90 links ← Gets complex quickly!

- ✓ Best for: Small number of VNets (< 5)
- ✗ Not scalable for large environments

Solution 2: Hub-Spoke with NVA (Recommended)

HUB-SPOKE TOPOLOGY WITH NVA:



Traffic Flow A → C:

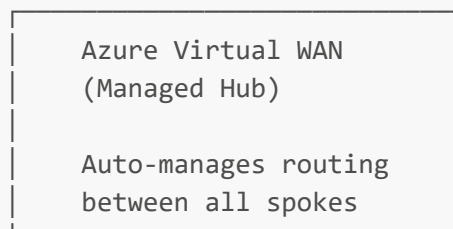
VNet-A → Hub NVA → VNet-C

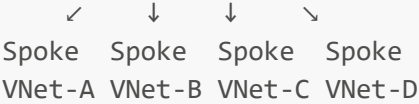
Requires:

1. UDR in each Spoke: Route traffic to Hub NVA
2. IP Forwarding enabled on NVA
3. NVA routes traffic to destination spoke

Solution 3: Azure Virtual WAN

AZURE VIRTUAL WAN (Managed Hub-Spoke):





- ✓ Fully managed by Azure
- ✓ Auto-routing between spokes
- ✓ Scales to hundreds of VNets
- ✓ Built-in VPN and ExpressRoute support

8.4 UDR Configuration for Hub-Spoke

USER-DEFINED ROUTES FOR HUB-SPOKE:

SPOKE-A Route Table (Applied to all Spoke-A subnets):

Destination	Next Hop Type	Next Hop IP
10.1.0.0/16	Virt Appliance	10.0.0.4 (Hub NVA)
10.2.0.0/16	Virt Appliance	10.0.0.4 (Hub NVA)
10.3.0.0/16	Virt Appliance	10.0.0.4 (Hub NVA)

HUB NVA must have:

- IP Forwarding ENABLED on the NIC
- Routing rules to forward to correct spoke
- (Or Azure Firewall handles this automatically)

Section 9: 2-Tier Application Architecture

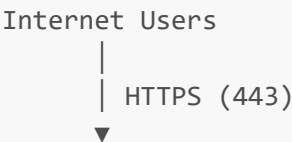
9.1 What is a 2-Tier Architecture?

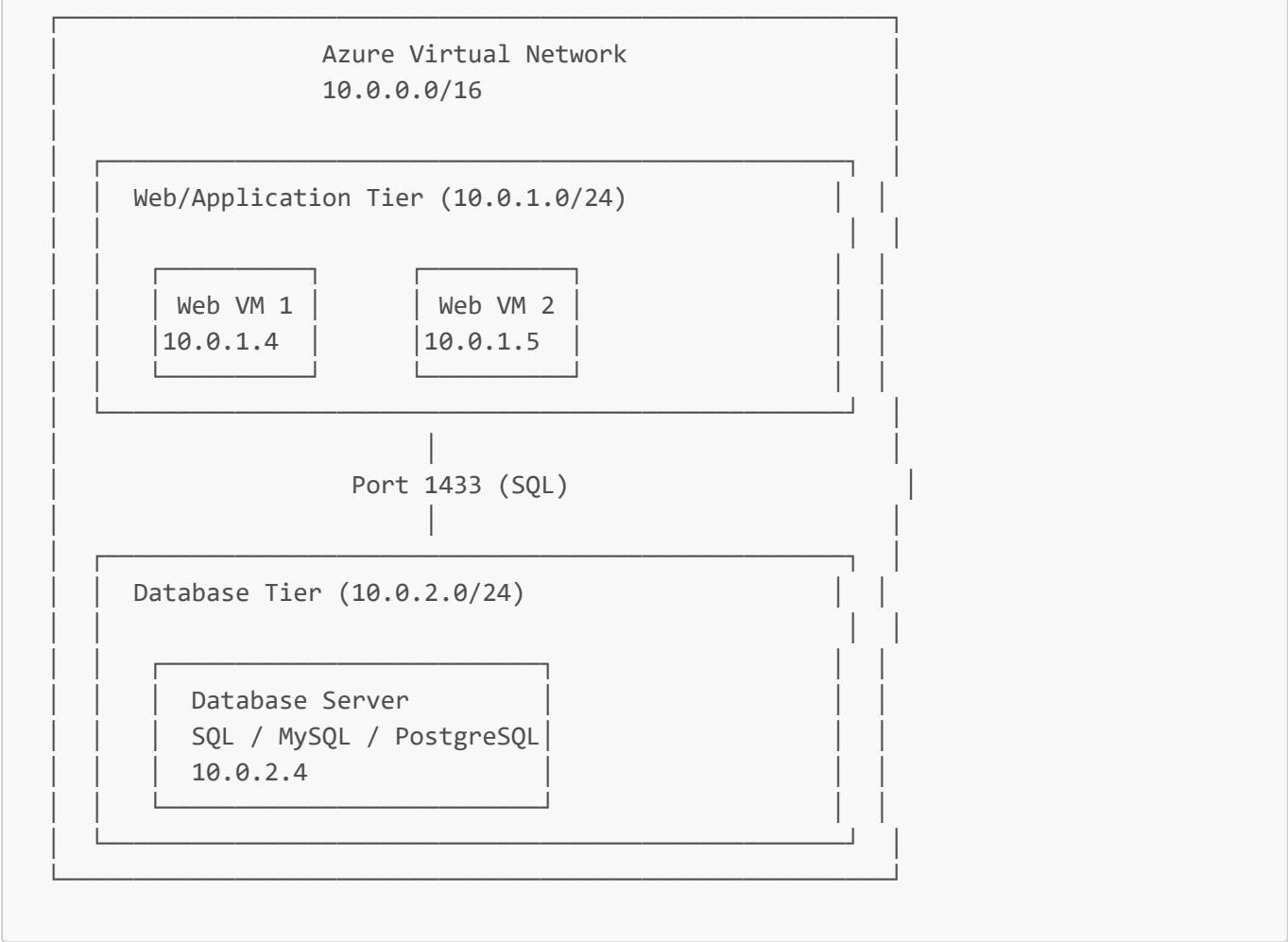
A **2-tier architecture** (also called client-server architecture) divides an application into two layers:

1. **Presentation/Web Tier** — User-facing layer (web servers, frontend)
2. **Data Tier** — Database layer (data storage and retrieval)

9.2 2-Tier Architecture Overview

2-TIER APPLICATION ARCHITECTURE:





9.3 2-Tier IP Address Planning

2-TIER IP ADDRESS PLAN:

Virtual Network: 10.0.0.0/16

Tier	Subnet CIDR	VM IPs
Web/App Tier	10.0.1.0/24	10.0.1.4 - .254 Web-VM1: 10.0.1.4 Web-VM2: 10.0.1.5
Database Tier	10.0.2.0/24	10.0.2.4 - .254 DB-Server: 10.0.2.4

9.4 2-Tier NSG Design

WEB TIER NSG RULES:

INBOUND:

Priority	Source	Destination	Port	Action
100	Internet	10.0.1.0/24	80	Allow
110	Internet	10.0.1.0/24	443	Allow
120	[AdminIP]	10.0.1.0/24	22	Allow
65500	*	*	*	Deny

OUTBOUND:

Priority	Source	Destination	Port	Action
100	10.0.1.0/24	10.0.2.0/24	1433	Allow
200	10.0.1.0/24	Internet	443	Allow
65500	*	*	*	Deny

DATABASE TIER NSG RULES:

INBOUND:

Priority	Source	Destination	Port	Action
100	10.0.1.0/24	10.0.2.0/24	1433	Allow
65500	*	*	*	Deny

OUTBOUND:

Priority	Source	Destination	Port	Action
100	10.0.2.0/24	10.0.1.0/24	*	Allow
65500	*	*	*	Deny

9.5 2-Tier Traffic Flow

2-TIER REQUEST FLOW:

User Browser → HTTPS Request → www.example.com

Step 1: DNS resolves to Load Balancer Public IP
User → DNS → 20.50.100.1 (Public IP)

Step 2: Load Balancer distributes to Web VM
20.50.100.1 → 10.0.1.4 (Web-VM1)
NSG Check: Port 443 from Internet = ALLOW 

Step 3: Web VM processes request, queries database
10.0.1.4 → 10.0.2.4 (DB-Server) Port 1433
NSG Check: Port 1433 from Web Tier = ALLOW ✓

Step 4: Database returns data to Web VM
10.0.2.4 → 10.0.1.4 (Response)

Step 5: Web VM sends HTML response to user
10.0.1.4 → User Browser

Section 10: 3-Tier Application Architecture

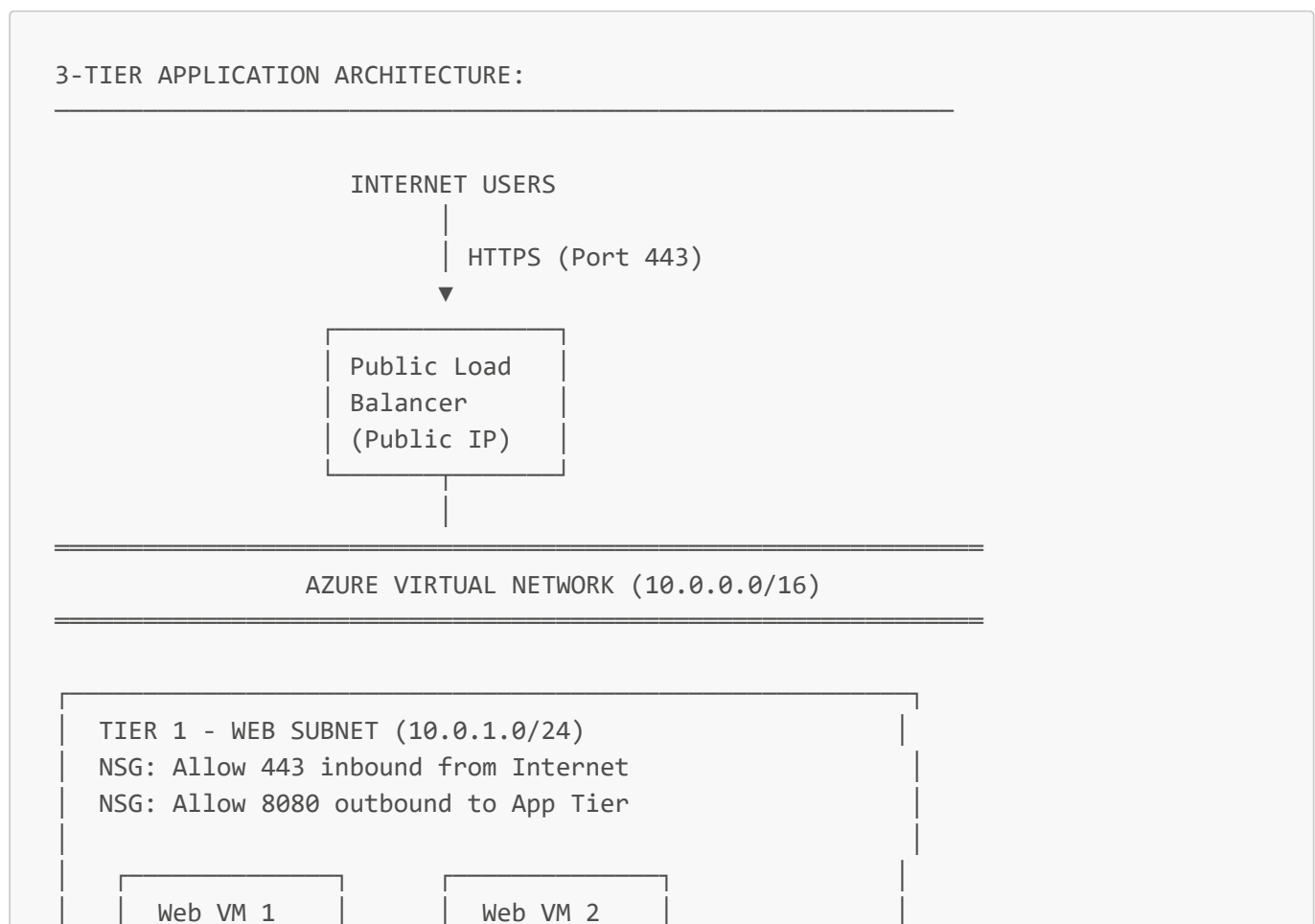
10.1 What is a 3-Tier Architecture?

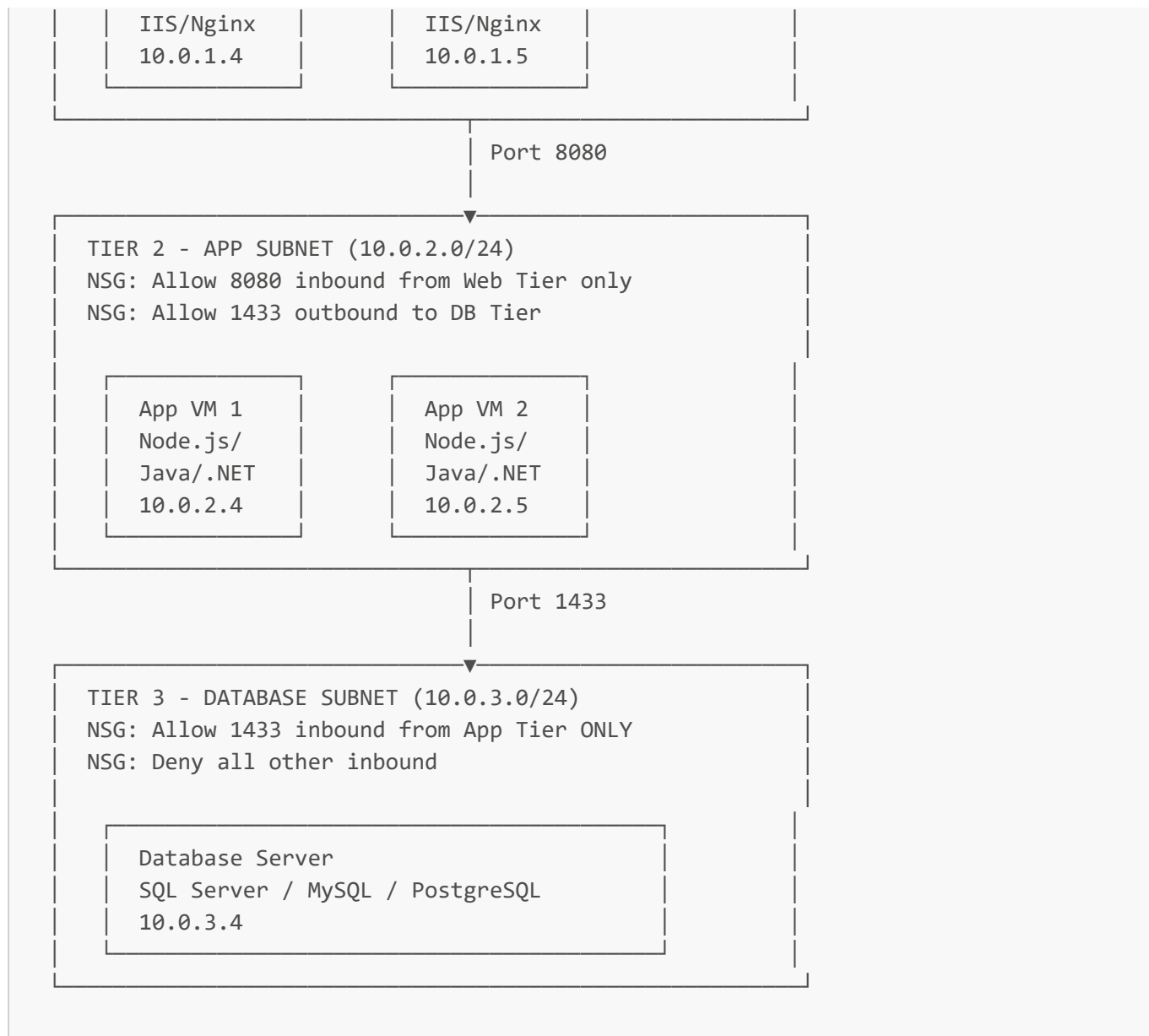
A **3-tier architecture** extends the 2-tier model by separating the application logic into its own dedicated tier:

1. **Presentation Tier (Web)** — User interface, HTTP/HTTPS handling
2. **Application/Logic Tier (App)** — Business logic, API processing
3. **Data Tier (Database)** — Data persistence and retrieval

This separation provides better **security**, **scalability**, and **maintainability**.

10.2 3-Tier Architecture Overview





10.3 3-Tier IP Address Plan

3-TIER IP ADDRESS PLANNING:

Virtual Network: 10.0.0.0/16

Tier	Subnet	Size	VM IPs
Web	10.0.1.0/24	/24	.4, .5, .6
Application	10.0.2.0/24	/24	.4, .5, .6
Database	10.0.3.0/24	/24	.4 (Primary)
Management	10.0.4.0/28	/28	.4 (Bastion)
Gateway	10.0.5.0/27	/27	Reserved

Note: GatewaySubnet MUST be named exactly "GatewaySubnet"

10.4 3-Tier NSG Design

WEB TIER NSG:

INBOUND RULES:

Priority	Source	Destination	Port	Action
100	Internet	10.0.1.0/24	80	Allow
110	Internet	10.0.1.0/24	443	Allow
120	AzureLB	10.0.1.0/24	*	Allow
65500	*	*	*	Deny

OUTBOUND RULES:

Priority	Source	Destination	Port	Action
100	10.0.1.0/24	10.0.2.0/24	8080	Allow
200	10.0.1.0/24	Internet	443	Allow
65500	*	*	*	Deny

APP TIER NSG:

INBOUND RULES:

Priority	Source	Destination	Port	Action
100	10.0.1.0/24	10.0.2.0/24	8080	Allow
65500	*	*	*	Deny

OUTBOUND RULES:

Priority	Source	Destination	Port	Action
100	10.0.2.0/24	10.0.3.0/24	1433	Allow
200	10.0.2.0/24	Internet	443	Allow
65500	*	*	*	Deny

DB TIER NSG:

INBOUND RULES:

Priority	Source	Destination	Port	Action
100	10.0.2.0/24	10.0.3.0/24	1433	Allow

65500	*	*	*	Deny
-------	---	---	---	------

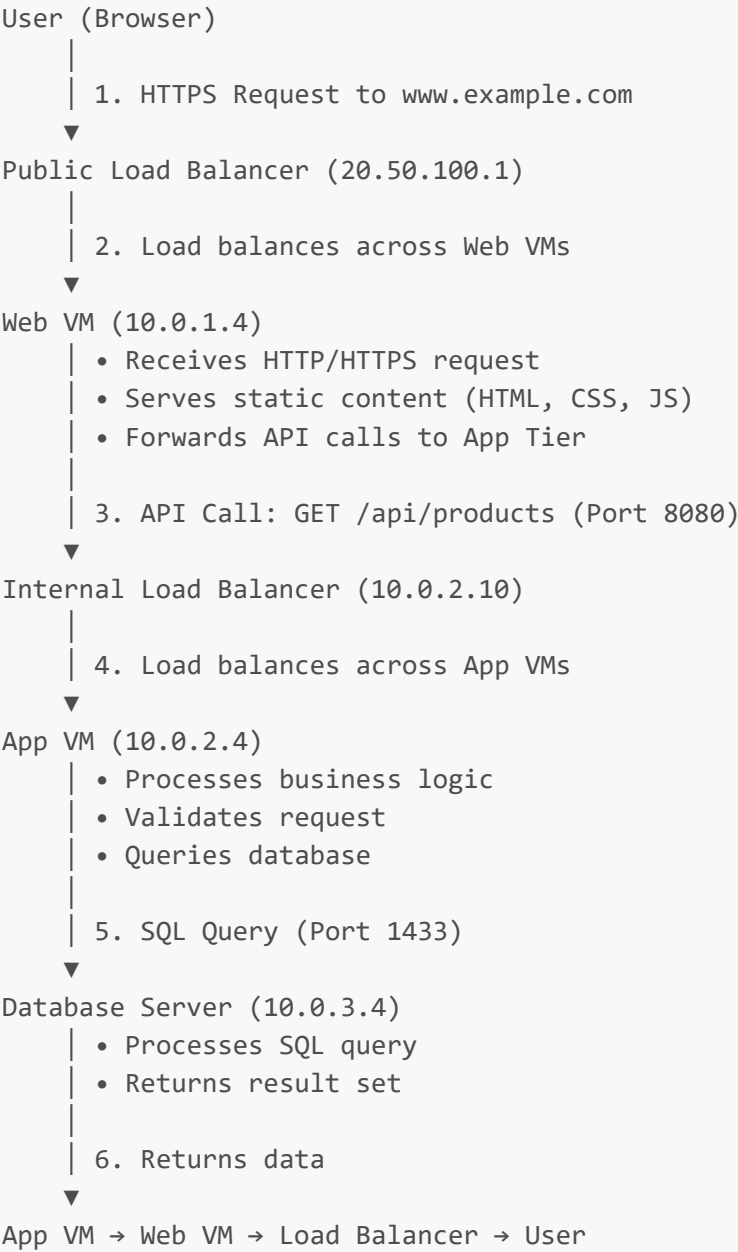
OUTBOUND RULES:

Priority	Source	Destination	Port	Action
100	10.0.3.0/24	10.0.2.0/24	*	Allow
65500	*	*	*	Deny

10.5 3-Tier Traffic Flow

3-TIER REQUEST FLOW (User Accessing Web App):

REQUEST JOURNEY:

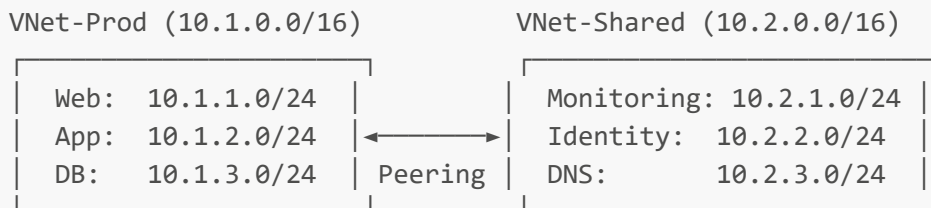


RESPONSE JOURNEY (Reverse Path):
DB → App VM → Internal LB → Web VM → Public LB → User

10.6 3-Tier with VNet Peering (Multi-VNet)

3-TIER ACROSS MULTIPLE VNets:

Use Case: Separate shared services into dedicated VNet



Peering allows:

- Production VMs to use Shared DNS
- Monitoring tools to access Production metrics
- Identity services to authenticate users

Section 11: Implementation Process

Overview

This section provides step-by-step guidance for implementing:

1. A 3-Tier Application in a single VNet
2. VNet Peering between two VNets

All steps use the Azure Portal (GUI).

11.1 Implementation: 3-Tier Application in Single VNet

Prerequisites

Before starting, ensure you have:

- ✓ Azure Subscription with Contributor access
- ✓ Resource Group created or permission to create one
- ✓ Understanding of your IP address plan
- ✓ Azure Portal access: portal.azure.com

IP ADDRESS PLAN FOR THIS IMPLEMENTATION:

Resource	Value
Resource Group	RG-ThreeTier-Demo
Region	East US
VNet Name	VNet-ThreeTier
VNet Address Space	10.0.0.0/16
Web Subnet	10.0.1.0/24
App Subnet	10.0.2.0/24
DB Subnet	10.0.3.0/24

STEP 1: Create Resource Group

PURPOSE: Logical container for all related resources

NAVIGATION:

Portal → Home → Resource Groups → + Create

CONFIGURATION:

Field	Value
Subscription	[Your Subscription]
Resource Group Name	RG-ThreeTier-Demo
Region	East US

STEPS IN PORTAL:

1. Click "Home" in left menu
2. Click "Resource Groups"
3. Click "+ Create" button
4. Select your Subscription
5. Enter Name: RG-ThreeTier-Demo
6. Select Region: East US
7. Click "Review + Create"
8. Click "Create"

STEP 2: Create Virtual Network with Subnets

PURPOSE: Create the network foundation with three subnets

NAVIGATION:

Portal → Search "Virtual Networks" → + Create

TAB 1 - BASICS:

Field	Value
Subscription	[Your Subscription]
Resource Group	RG-ThreeTier-Demo
Name	VNet-ThreeTier
Region	East US

TAB 2 - IP ADDRESSES:

Address Space Configuration:

IPv4 Address Space	10.0.0.0/16
--------------------	-------------

Delete the default subnet and create three new subnets:

SUBNET 1 - Web Tier:

Subnet Name	Subnet-Web
Subnet Address Range	10.0.1.0/24

SUBNET 2 - App Tier:

Subnet Name	Subnet-App
Subnet Address Range	10.0.2.0/24

SUBNET 3 - Database Tier:

Subnet Name	Subnet-DB
Subnet Address Range	10.0.3.0/24

STEPS IN PORTAL:

1. Search for "Virtual Networks" in Azure Portal
2. Click "+ Create"
3. Fill in Basics tab
4. Navigate to "IP Addresses" tab
5. Click on "default" subnet → Delete it
6. Click "+ Add subnet" → Create Subnet-Web (10.0.1.0/24)
7. Click "+ Add subnet" → Create Subnet-App (10.0.2.0/24)
8. Click "+ Add subnet" → Create Subnet-DB (10.0.3.0/24)
9. Click "Review + Create" → "Create"

STEP 3: Create Network Security Groups

PURPOSE: Define firewall rules for each tier

NAVIGATION:

Portal → Search "Network Security Groups" → + Create

CREATE NSG 1 - WEB TIER:

Field	Value
Resource Group	RG-ThreeTier-Demo
Name	NSG-Web
Region	East US

After creating NSG-Web, add inbound rules:

INBOUND RULE 1 - Allow HTTP:

Source	Service Tag
Source Service Tag	Internet
Source Port Ranges	*
Destination	Any
Service	HTTP
Destination Port	80
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-HTTP-Inbound

INBOUND RULE 2 - Allow HTTPS:

Source	Service Tag
Source Service Tag	Internet
Source Port Ranges	*
Destination	Any
Service	HTTPS
Destination Port	443
Protocol	TCP
Action	Allow
Priority	110
Name	Allow-HTTPS-Inbound

OUTBOUND RULE 1 - Allow to App Tier:

Source	IP Addresses
Source IP Ranges	10.0.1.0/24
Source Port Ranges	*
Destination	IP Addresses
Destination IP	10.0.2.0/24

Destination Port	8080
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-To-AppTier

CREATE NSG 2 - APP TIER:

Name	NSG-App
Region	East US
Resource Group	RG-ThreeTier-Demo

INBOUND RULE - Allow from Web Tier only:

Source	IP Addresses
Source IP Ranges	10.0.1.0/24
Source Port Ranges	*
Destination	Any
Destination Port	8080
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-From-WebTier

OUTBOUND RULE - Allow to DB Tier:

Source	IP Addresses
Source IP Ranges	10.0.2.0/24
Source Port Ranges	*
Destination	IP Addresses
Destination IP	10.0.3.0/24
Destination Port	1433
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-To-DBTier

CREATE NSG 3 - DB TIER:

Name	NSG-DB
Region	East US
Resource Group	RG-ThreeTier-Demo

INBOUND RULE - Allow from App Tier only:

Source	IP Addresses
Source IP Ranges	10.0.2.0/24
Source Port Ranges	*
Destination	Any
Destination Port	1433
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-From-AppTier-SQL

STEP 4: Associate NSGs with Subnets

PURPOSE: Apply firewall rules to each subnet

NAVIGATION:

Portal → NSG-Web → Settings → Subnets → + Associate

ASSOCIATE NSG-Web WITH Subnet-Web:

1. Go to "Network Security Groups"
2. Click on "NSG-Web"
3. In left menu, click "Subnets" under Settings
4. Click "+ Associate"
5. Select:
Virtual Network: VNet-ThreeTier
Subnet: Subnet-Web
6. Click "OK"

ASSOCIATE NSG-App WITH Subnet-App:

1. Go to "Network Security Groups"
2. Click on "NSG-App"
3. Click "Subnets" → "+ Associate"
4. Select:
Virtual Network: VNet-ThreeTier
Subnet: Subnet-App
5. Click "OK"

ASSOCIATE NSG-DB WITH Subnet-DB:

1. Go to "Network Security Groups"
2. Click on "NSG-DB"
3. Click "Subnets" → "+ Associate"
4. Select:

Virtual Network: VNet-ThreeTier
Subnet: Subnet-DB
5. Click "OK"

VERIFICATION:

Go to VNet-ThreeTier → Subnets

Each subnet should show its associated NSG:

- ✓ Subnet-Web → NSG-Web
- ✓ Subnet-App → NSG-App
- ✓ Subnet-DB → NSG-DB

STEP 5: Deploy Web Tier Virtual Machines

PURPOSE: Create web servers in the Web Tier subnet

NAVIGATION:

Portal → Virtual Machines → + Create → Azure Virtual Machine

VM CONFIGURATION - Web VM 1:

BASICS TAB:

Field	Value
Subscription	[Your Subscription]
Resource Group	RG-ThreeTier-Demo
VM Name	VM-Web-01
Region	East US
Availability Options	Availability Zone
Zone	Zone 1
Image	Windows Server 2022
Size	Standard_B2s
Username	azureadmin
Password	[Strong Password]
Public Inbound Ports	None (NSG will handle it)

NETWORKING TAB:

Field	Value
Virtual Network	VNet-ThreeTier
Subnet	Subnet-Web (10.0.1.0/24)
Public IP	None
NIC NSG	None (Using Subnet NSG)
Load Balancing	None (configure separately)

1. Click "Review + Create"
2. Validate configuration
3. Click "Create"
4. Wait for deployment (~3-5 minutes)

REPEAT FOR Web VM 2:

- Name: VM-Web-02
- Zone: Zone 2 (for high availability)
- Same subnet: Subnet-Web

STEP 6: Deploy Application Tier Virtual Machines

PURPOSE: Create application servers in App Tier subnet

VM CONFIGURATION - App VM 1:

BASICS TAB:

Field	Value
Resource Group	RG-ThreeTier-Demo
VM Name	VM-App-01
Region	East US
Availability Zone	Zone 1
Image	Windows Server 2022
Size	Standard_B2s
Username	azureadmin
Password	[Strong Password]
Public Inbound Ports	None

NETWORKING TAB:

Virtual Network	VNet-ThreeTier
Subnet	Subnet-App (10.0.2.0/24)
Public IP	None
NIC NSG	None

REPEAT FOR App VM 2:

- Name: VM-App-02
- Zone: Zone 2
- Same subnet: Subnet-App

STEP 7: Deploy Database Tier

PURPOSE: Create database server in DB Tier subnet

OPTION A: Azure SQL Database (Recommended for PaaS)

NAVIGATION:

Portal → Search "SQL Databases" → + Create

CONFIGURATION:

Field	Value
Resource Group	RG-ThreeTier-Demo
Database Name	DB-ThreeTier
Server	Create New
Server Name	sql-threetier-demo
Authentication	SQL Authentication
Admin Login	sqladmin
Password	[Strong Password]
Compute + Storage	General Purpose - Standard

NETWORKING TAB:

Connectivity Method	Private Endpoint
Virtual Network	VNet-ThreeTier
Subnet	Subnet-DB

OPTION B: SQL Server on Virtual Machine (IaaS)

VM CONFIGURATION - DB Server:

VM Name	VM-DB-01
Image	SQL Server 2022 on Win2022
Size	Standard_D4s_v3
Subnet	Subnet-DB (10.0.3.0/24)
Public IP	None

STEP 8: Create Load Balancer (Public - Web Tier)

PURPOSE: Distribute internet traffic across Web VMs

NAVIGATION:

Portal → Search "Load Balancers" → + Create

TAB 1 - BASICS:

Field	Value
Resource Group	RG-ThreeTier-Demo
Name	LB-Web-Public
Region	East US
SKU	Standard
Type	Public
Tier	Regional

TAB 2 - FRONTEND IP CONFIGURATION:

Click "+ Add a frontend IP configuration"

Name	Frontend-Web
IP Version	IPv4
IP Type	IP Address
Public IP Address	Create New
Public IP Name	PIP-Web-LB
Availability Zone	Zone-Redundant

TAB 3 - BACKEND POOLS:

Click "+ Add a backend pool"

Name	BackendPool-Web
Virtual Network	VNet-ThreeTier
Backend Pool Config	NIC

Add VMs to backend pool:

- VM-Web-01 → NIC → IP Configuration
- VM-Web-02 → NIC → IP Configuration

TAB 4 - INBOUND RULES:

Click "+ Add a load balancing rule"

Name	Rule-HTTP
Frontend IP	Frontend-Web
Backend Pool	BackendPool-Web
Protocol	TCP
Port	80
Backend Port	80
Health Probe	Create New
Health Probe Name	Probe-HTTP
Health Probe Protocol	HTTP
Health Probe Port	80

Health Probe Path	/
Session Persistence	None

Add second rule for HTTPS:

Name	Rule-HTTPS
Port	443
Backend Port	443

Click "Review + Create" → "Create"

STEP 9: Create Internal Load Balancer (App Tier)

PURPOSE: Distribute traffic from Web Tier to App VMs

NAVIGATION:

Portal → Load Balancers → + Create

BASICS:

Name	LB-App-Internal
SKU	Standard
Type	Internal

FRONTEND IP:

Name	Frontend-App
Virtual Network	VNet-ThreeTier
Subnet	Subnet-App
IP Assignment	Static
IP Address	10.0.2.10

BACKEND POOL:

Name	BackendPool-App
Add VMs	VM-App-01, VM-App-02

LOAD BALANCING RULE:

Name	Rule-AppAPI
Frontend IP	Frontend-App (10.0.2.10)

Backend Pool	BackendPool-App
Protocol	TCP
Port	8080
Backend Port	8080

STEP 10: Verify 3-Tier Connectivity

VERIFICATION STEPS:

STEP 10.1: Verify NSG Rules Applied

1. Go to VNet-ThreeTier → Subnets
2. Click each subnet → Verify NSG is associated
3. Check: Subnet-Web shows NSG-Web 
4. Check: Subnet-App shows NSG-App 
5. Check: Subnet-DB shows NSG-DB 

STEP 10.2: Test Connectivity Using Azure Network Watcher

1. Search "Network Watcher" in portal
2. Click "Connection Monitor" → "+ Create"
3. Test 1: Web VM → App VM on Port 8080 (should succeed)
4. Test 2: Web VM → DB VM on Port 1433 (should fail - NSG blocks)
5. Test 3: App VM → DB VM on Port 1433 (should succeed)

STEP 10.3: Verify IP Flow

1. Network Watcher → IP Flow Verify
2. Test: Internet → Web VM Port 443 → Should: ALLOW 
3. Test: Internet → App VM Port 8080 → Should: DENY 
4. Test: Internet → DB VM Port 1433 → Should: DENY 

EXPECTED RESULTS:

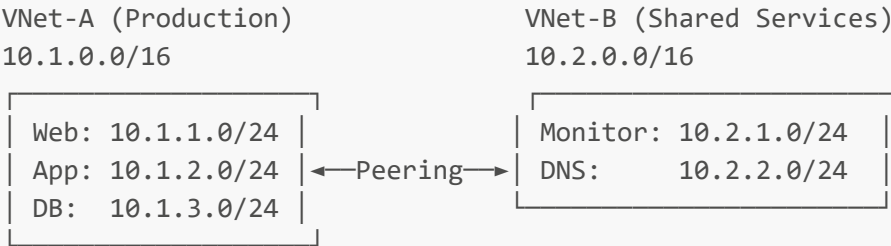
Test	Result
Internet → Web VM (Port 443)	 Allow
Internet → App VM (Port 8080)	 Deny
Internet → DB VM (Port 1433)	 Deny
Web VM → App VM (Port 8080)	 Allow
Web VM → DB VM (Port 1433)	 Deny
App VM → DB VM (Port 1433)	 Allow
DB VM → Internet (Any)	 Deny

11.2 Implementation: VNet Peering

Scenario Overview

VNet PEERING SCENARIO:

We will create two VNets in the same region and peer them:



GOAL: VNet-A resources can communicate with VNet-B services

STEP 1: Create VNet-A (Production)

NAVIGATION:
Portal → Virtual Networks → + Create

CONFIGURATION VNet-A:

BASICS TAB:

Resource Group	RG-Peering-Demo
Name	VNet-A-Production
Region	East US

IP ADDRESSES TAB:

IPv4 Address Space	10.1.0.0/16
--------------------	-------------

SUBNETS:

Subnet 1 Name	Subnet-Web-A
Subnet 1 Range	10.1.1.0/24
Subnet 2 Name	Subnet-App-A
Subnet 2 Range	10.1.2.0/24

Click "Review + Create" → "Create"

STEP 2: Create VNet-B (Shared Services)

NAVIGATION:
Portal → Virtual Networks → + Create

CONFIGURATION VNet-B:

BASICS TAB:

Resource Group	RG-Peering-Demo
Name	VNet-B-Shared
Region	East US

IP ADDRESSES TAB:

IPv4 Address Space	10.2.0.0/16
--------------------	-------------

SUBNETS:

Subnet 1 Name	Subnet-Monitor
Subnet 1 Range	10.2.1.0/24
Subnet 2 Name	Subnet-DNS
Subnet 2 Range	10.2.2.0/24

Click "Review + Create" → "Create"

STEP 3: Deploy Test Virtual Machines

PURPOSE: Create VMs to test peering connectivity

VM IN VNet-A (Production):

Name	VM-Prod-01
Resource Group	RG-Peering-Demo
Region	East US
Image	Ubuntu Server 22.04 LTS
Size	Standard_B1s
VNet	VNet-A-Production
Subnet	Subnet-Web-A (10.1.1.0/24)
Public IP	None

VM IN VNet-B (Shared Services):

--	--

Name	VM-Shared-01
Resource Group	RG-Peering-Demo
Region	East US
Image	Ubuntu Server 22.04 LTS
Size	Standard_B1s
VNet	VNet-B-Shared
Subnet	Subnet-Monitor (10.2.1.0/24)
Public IP	None

NOTE THE PRIVATE IP ADDRESSES:

VM-Prod-01: Should be 10.1.1.4

VM-Shared-01: Should be 10.2.1.4

STEP 4: Configure VNet Peering (VNet-A to VNet-B)

IMPORTANT: Peering requires TWO connections:

Connection 1: VNet-A → VNet-B

Connection 2: VNet-B → VNet-A

NAVIGATION:

Portal → Virtual Networks → VNet-A-Production → Peerings

CREATE PEERING FROM VNet-A TO VNet-B:

1. Go to VNet-A-Production
2. In left menu, scroll to "Settings" → Click "Peerings"
3. Click "+ Add"

PEERING CONFIGURATION FORM:

This Virtual Network (VNet-A) Section:

Field	Value
Peering Link Name	VNetA-to-VNetB
Traffic to remote VNet	Allow
Traffic from remote VNet	Allow
Virtual Network Gateway	None

Remote Virtual Network (VNet-B) Section:

Field	Value
Peering Link Name	VNetB-to-VNetA
Virtual Network Deployment	Resource Manager
Subscription	[Your Subscription]
Virtual Network	VNet-B-Shared

Traffic to remote VNet	Allow
Traffic from remote VNet	Allow
Virtual Network Gateway	None

4. Click "Add"
5. Wait for peering status to show "Connected"

VERIFY PEERING STATUS:

After creation, you should see in VNet-A Peerings:

Peering Name	Status	Remote VNet
VNetA-to-VNetB	Connected 	VNet-B-Shared

And in VNet-B Peerings:

Peering Name	Status	Remote VNet
VNetB-to-VNetA	Connected 	VNet-A-Production

STATUS MEANINGS:

- Initiated: One side created, waiting for other side
- Connected: Both sides created and working 
- Disconnected: Problem - check configuration 

STEP 5: Verify Peering - Routing Table Check

VERIFY ROUTING TABLES UPDATED:

NAVIGATION:

Portal → VNet-A-Production → Subnets → Subnet-Web-A
→ View Route Table

OR

Portal → Network Watcher → Effective Routes
→ Select VM-Prod-01 NIC

Expected Effective Routes on VM-Prod-01 (VNet-A):

Destination	Next Hop
10.1.0.0/16	Virtual Network (local)
10.2.0.0/16	VNet Peering ← SHOULD APPEAR AFTER PEER

0.0.0.0/0	Internet
-----------	----------

If 10.2.0.0/16 with "VNet Peering" next hop appears =  SUCCESS

STEP 6: Test Peering Connectivity

TEST METHOD 1: Azure Network Watcher - Connection Monitor

NAVIGATION:

Portal → Network Watcher → Connection Monitor → + Create

CONFIGURATION:

Name	Test-VNetA-to-VNetB
Region	East US

SOURCE:

Source Type	Virtual Machine
Virtual Machine	VM-Prod-01 (in VNet-A)

DESTINATION + TEST:

Destination Type	Virtual Machine
Destination VM	VM-Shared-01 (in VNet-B)
Port	22 (SSH) or ICMP

Click "Create" → Wait for test results

Expected: Connection successful 

TEST METHOD 2: Azure Network Watcher - IP Flow Verify

NAVIGATION:

Portal → Network Watcher → IP Flow Verify

Virtual Machine	VM-Prod-01
Direction	Outbound
Protocol	TCP
Local Port	60000
Remote IP	10.2.1.4 (VM-Shared-01 IP)
Remote Port	22

Expected: "Access allowed" 

EXPECTED TEST RESULTS:

Test	Result
VM-Prod-01 → VM-Shared-01 (Peered)	
VM-Shared-01 → VM-Prod-01 (Peered)	
VM-Prod-01 → 10.3.0.4 (Non-Peered VNet)	

STEP 7: Configure NSGs for Peered Traffic

PURPOSE: Allow specific traffic between peered VNets

SCENARIO: Allow monitoring VM (VNet-B) to ping/monitor VNet-A VMs

ADD NSG RULE TO VNet-A SUBNET NSG:

NAVIGATION:

Portal → NSG (associated with Subnet-Web-A) →
Inbound Security Rules → + Add

RULE CONFIGURATION:

Source	IP Addresses
Source IP Ranges	10.2.0.0/16
Source Port Ranges	*
Destination	IP Addresses
Destination IP	10.1.0.0/16
Destination Port	*
Protocol	ICMP
Action	Allow
Priority	200
Name	Allow-From-SharedVNet-Ping

NOTE: Without this rule, even with peering, ICMP would
be blocked by the Deny-All-Inbound default rule

IMPORTANT CONCEPT:

VNet Peering enables ROUTING between VNets
NSGs control what TRAFFIC is actually allowed

Both must be configured correctly!

Peering = Opens the "road" between VNets
NSG = The "traffic rules" on that road

11.3 Implementation: Hub-Spoke with Transitive Routing

Scenario Overview

HUB-SPOKE SCENARIO:

```
Hub VNet (10.0.0.0/16)
├── Hub Subnet: 10.0.1.0/24 (Contains Azure Firewall/NVA)
├── Peering to Spoke-A VNet (10.1.0.0/16)
│   └── Spoke-A Subnet: 10.1.1.0/24
└── Peering to Spoke-B VNet (10.2.0.0/16)
    └── Spoke-B Subnet: 10.2.1.0/24
```

GOAL: Spoke-A VMs can communicate with Spoke-B VMs
through the Hub (transitive routing)

STEP 1: Create Hub VNet

NAVIGATION:

Portal → Virtual Networks → + Create

CONFIGURATION:

Name	VNet-Hub
Resource Group	RG-HubSpoke-Demo
Region	East US
Address Space	10.0.0.0/16

SUBNETS:

Hub Subnet	10.0.1.0/24
AzureFirewallSubnet	10.0.2.0/26 (Required name!)

IMPORTANT: The Azure Firewall subnet MUST be named
"AzureFirewallSubnet" exactly

STEP 2: Create Spoke VNets

CREATE VNet-Spoke-A:

Name	VNet-Spoke-A
Address Space	10.1.0.0/16
Subnet Name	Subnet-Spoke-A
Subnet Range	10.1.1.0/24

CREATE VNet-Spoke-B:

Name	VNet-Spoke-B
Address Space	10.2.0.0/16
Subnet Name	Subnet-Spoke-B
Subnet Range	10.2.1.0/24

STEP 3: Deploy Azure Firewall in Hub

PURPOSE: Azure Firewall will act as the NVA for routing between spokes

NAVIGATION:

Portal → Search "Firewalls" → + Create

CONFIGURATION:

Field	Value
Resource Group	RG-HubSpoke-Demo
Name	AzFirewall-Hub
Region	East US
Availability Zone	None
Firewall SKU	Standard
Firewall Policy	Create New: FWPolicy-Hub
Virtual Network	VNet-Hub
Public IP Address	Create New: PIP-Firewall

Click "Review + Create" → "Create"

NOTE: Azure Firewall takes ~5-10 minutes to deploy

RECORD THE FIREWALL PRIVATE IP:

After deployment → Click AzFirewall-Hub →

Overview → Note "Private IP Address"

Example: 10.0.2.4 (This will be your NVA IP for routing)

STEP 4: Create Hub-Spoke Peerings

PEERING 1: Hub ↔ Spoke-A

NAVIGATION:

Portal → VNet-Hub → Peerings → + Add

This VNet (Hub) Configuration:

Peering Link Name	Hub-to-SpokeA
Traffic to remote	Allow
Traffic from remote	Allow
Gateway Transit	Allow gateway transit: OFF

Remote VNet (Spoke-A) Configuration:

Peering Link Name	SpokeA-to-Hub
Virtual Network	VNet-Spoke-A
Traffic to remote	Allow
Traffic from remote	Allow
Use Remote Gateway	OFF (No gateway deployed)

Click "Add"

PEERING 2: Hub ↔ Spoke-B

NAVIGATION:

Portal → VNet-Hub → Peerings → + Add

This VNet (Hub) Configuration:

Peering Link Name	Hub-to-SpokeB
Traffic to remote	Allow
Traffic from remote	Allow

Remote VNet (Spoke-B) Configuration:

Peering Link Name	SpokeB-to-Hub
Virtual Network	VNet-Spoke-B
Traffic to remote	Allow

Traffic from remote	Allow
---------------------	-------

Click "Add"

STEP 5: Create User-Defined Routes (UDR) for Transitive Routing

PURPOSE: Force spoke-to-spoke traffic through Hub Firewall

CREATE ROUTE TABLE FOR SPOKE-A:

NAVIGATION:

Portal → Search "Route Tables" → + Create

BASICS:

Name	RT-SpokeA
Resource Group	RG-HubSpoke-Demo
Region	East US
Propagate gateway	Yes

Click "Create"

ADD ROUTES TO RT-SpokeA:

NAVIGATION:

Portal → RT-SpokeA → Settings → Routes → + Add

ROUTE 1 - Send Spoke-B traffic through Firewall:

Route Name	To-SpokeB-via-Firewall
Destination Type	IP Addresses
Destination CIDR	10.2.0.0/16
Next Hop Type	Virtual Appliance
Next Hop IP Address	10.0.2.4 (Firewall Private IP)

Click "Add"

ASSOCIATE RT-SpokeA WITH Subnet-Spoke-A:

NAVIGATION:

Portal → RT-SpokeA → Settings → Subnets → + Associate

Virtual Network	VNet-Spoke-A
-----------------	--------------

Subnet	Subnet-Spoke-A
--------	----------------

Click "OK"

CREATE ROUTE TABLE FOR SPOKE-B:

NAVIGATION:

Portal → Route Tables → + Create

BASICS:

Name	RT-SpokeB
------	-----------

ADD ROUTE:

Route Name	To-SpokeA-via-Firewall
Destination CIDR	10.1.0.0/16
Next Hop Type	Virtual Appliance
Next Hop IP	10.0.2.4 (Firewall Private IP)

ASSOCIATE WITH Subnet-Spoke-B:

Virtual Network	VNet-Spoke-B
Subnet	Subnet-Spoke-B

STEP 6: Configure Azure Firewall Policy Rules

PURPOSE: Allow traffic between spokes through firewall

NAVIGATION:

Portal → Firewall Policies → FWPolicy-Hub →
Network Rule Collections → + Add

NETWORK RULE COLLECTION:

Name	Allow-Spoke-to-Spoke
Rule Collection Type	Network
Priority	100
Action	Allow

ADD RULE:

Name	Allow-SpokeA-to-SpokeB
Source Type	IP Address
Source	10.1.0.0/16
Protocol	TCP,UDP,ICMP
Destination Ports	*
Destination Type	IP Address
Destination	10.2.0.0/16

ADD RULE 2:

Name	Allow-SpokeB-to-SpokeA
Source	10.2.0.0/16
Protocol	TCP,UDP,ICMP
Destination Ports	*
Destination	10.1.0.0/16

Click "Add"

STEP 7: Verify Hub-Spoke Transitive Routing

VERIFICATION STEPS:

STEP 7.1: Check Effective Routes on Spoke-A VM

Portal → VM in Spoke-A → Networking →
Network Interface → Effective Routes

Should see:

10.1.0.0/16	Virtual Network
10.0.0.0/16	VNet Peering (Hub)
10.2.0.0/16	User Defined → 10.0.2.4 (Firewall)

STEP 7.2: Test Spoke-A to Spoke-B via Network Watcher

Portal → Network Watcher → Connection Monitor

Source: VM-SpokeA (10.1.1.4)

Destination: VM-SpokeB (10.2.1.4)

Port: ICMP or 22

Expected: Connected  (Traffic routed through Firewall)

STEP 7.3: Verify in Azure Firewall Logs

Portal → AzFirewall-Hub → Monitoring → Logs

Check Network Rules Log for entries showing:

Source: 10.1.x.x → Destination: 10.2.x.x → Allow

TRAFFIC FLOW CONFIRMATION:

VM-SpokeA (10.1.1.4)

|

| UDR: 10.2.0.0/16 → 10.0.2.4

▼

Azure Firewall (10.0.2.4) in Hub

|

| Firewall Rule: Allow Spoke-to-Spoke

▼

VM-SpokeB (10.2.1.4)

✅ Connection Successful!

11.4 Troubleshooting Guide

COMMON ISSUES AND SOLUTIONS:

ISSUE 1: VNet Peering Status "Disconnected"

Symptoms: Peering shows "Disconnected" or "Initiated"

Cause: Only one side of the peering was created

Solution:

1. Go to both VNets → Peerings
2. Verify BOTH sides show "Connected"
3. If one shows "Initiated" → The other side is missing
4. Delete and recreate the peering from the other VNet

ISSUE 2: Cannot Reach Peered VNet VMs

Symptoms: Ping/connection fails despite peering being "Connected"

Cause: NSG blocking traffic from remote VNet

Solution:

1. Network Watcher → IP Flow Verify
2. Identify which NSG is blocking
3. Add inbound rule allowing source from peered VNet CIDR
4. Example: Allow from 10.2.0.0/16 on required port

ISSUE 3: Overlapping Address Spaces

Symptoms: Peering creation fails with CIDR conflict error

Cause: Both VNets have same or overlapping address ranges

Solution:

1. Check both VNet address spaces

2. One or both must be changed to non-overlapping ranges
3. May require deleting VMs and recreating VNets
4. Plan address spaces BEFORE deployment!

ISSUE 4: Hub-Spoke Traffic Not Flowing

Symptoms: Spoke-A cannot reach Spoke-B through Hub

Cause: Missing UDR, Firewall rule, or IP Forwarding

CHECKLIST:

- ☐ UDR created for each Spoke pointing to NVA IP
- ☐ UDR associated with correct Subnet
- ☐ NVA has IP Forwarding enabled on NIC
- ☐ Firewall/NVA has rules to allow spoke traffic
- ☐ Peering configured from both Hub→Spoke directions

Check IP Forwarding on Firewall NIC:

Portal → NVA VM → Networking → NIC → IP Configurations

- ☐ Enable IP Forwarding = Yes

ISSUE 5: NSG Rule Not Working

Symptoms: Traffic blocked despite having allow rule

Cause: Priority conflict or wrong direction

Solution:

1. Check Priority: Lower number = higher priority
2. Verify Direction: Inbound vs Outbound
3. Check if Deny rule has lower priority number
4. Verify Source and Destination IP/CIDR are correct
5. Use "Effective Security Rules" to see applied rules:
VM → Networking → NIC → Effective Security Rules

DIAGNOSTIC TOOLS:

Tool	Use Case
IP Flow Verify	Test if traffic allowed
Next Hop	Verify routing path
Connection Monitor	Test end-to-end connect
Effective Security Rules	See all applied NSG rules
Effective Routes	See all routing entries
VPN Troubleshoot	Debug VPN gateway issues

Summary and Reference

Architecture Decision Guide

CHOOSING THE RIGHT ARCHITECTURE:

SCENARIO → RECOMMENDATION

Simple web + database?

→ 2-Tier in single VNet

Web + API + Database?

→ 3-Tier in single VNet

Multiple teams/environments needing isolation?

→ Multiple VNets with Peering

Many VNets needing central management?

→ Hub-Spoke with Azure Firewall

Multi-region with global reach?

→ Azure Virtual WAN

On-premises to cloud connectivity?

→ Hub VNet with VPN Gateway or ExpressRoute

Quick Reference Card

QUICK REFERENCE	
CONCEPT	KEY POINTS
OSI Layer 3	IP Addressing, Routing
OSI Layer 4	TCP/UDP, Port Numbers
VLAN	L2 Segmentation (→ Subnets in Az)
VNet	Azure's private network
Subnet	Subdivisions within VNet
NSG	L3/L4 firewall for Azure
VNet Peering	Connect two VNets privately
Peering = NOT Trans	A↔B, B↔C ≠ A↔C
Hub-Spoke	Central VNet routes all traffic
UDR	Override Azure default routes
Azure Firewall	Managed NVA for hub-spoke
2-Tier	Web + Database
3-Tier	Web + App + Database
Azure Reserved IPs	5 per subnet (.0,.1,.2,.3,.255)
Peering Requirement	Non-overlapping CIDRs
Peering Direction	Both sides must be configured
NSG Processing	Subnet NSG → NIC NSG (Inbound)

NSG Priority	100-4096 (lower = higher priority)
--------------	------------------------------------

Documentation Version: 1.0
Last Updated: 2024
Platform: Microsoft Azure
Difficulty: Intermediate